In [2]:

```
import numpy as np

# Create a random array
array = np.random.randint(1, 100, 9)
print("Original Array:")
print(array)

# Square root of array elements
print("\nSquare Root:")
print(np.sqrt(array))

# Check dimensions
print("\nNumber of Dimensions:")
print(array.ndim)

# Reshape into 3x3 matrix
new_array = array.reshape(3, 3)
print("\nReshaped Array:")
print(new_array)

# Check dimensions of 2D array
print("\nDimensions of New Array:")
print(new_array.ndim)

# Convert 2D array into 1D array
print("\nRavel Array:")
print(new_array.ravel())

# Reshape again
newm = new_array.reshape(3, 3)
print("\nNew Matrix:")
print(newm)

# Array slicing
print("\nnewm[2, 1:3]:")
print(newm[2, 1:3])

print("\nnewm[1:2, 1:3]:")
print(newm[1:2, 1:3])

print("\nnew_array[0:3, 0:0]:")
print(new_array[0:3, 0:0])

print("\nnew_array[0:2, 0:1]:")
print(new_array[0:2, 0:1])

print("\nnew_array[0:3, 0:1]:")
print(new_array[0:3, 0:1])

print("\nnew_array[1:3]:")
print(new_array[1:3])
```

```
Original Array:
[32 12 27 32 30 45 48 10 52]

Square Root:
[5.65685425 3.46410162 5.19615242 5.65685425 5.47722558 6.70820393
 6.92820323 3.16227766 7.21110255]

Number of Dimensions:
1

Reshaped Array:
[[32 12 27]
 [32 30 45]
 [48 10 52]]

Dimensions of New Array:
2

Ravel Array:
[32 12 27 32 30 45 48 10 52]

New Matrix:
[[32 12 27]
 [32 30 45]
 [48 10 52]]

newm[2, 1:3]:
[10 52]

newm[1:2, 1:3]:
[[30 45]]

new_array[0:3, 0:0]:
[]

new_array[0:2, 0:1]:
[[32]
 [32]]

new_array[0:3, 0:1]:
[[32]
 [32]
 [48]]

new_array[1:3]:
[[32 30 45]
 [48 10 52]]
```

In [ ]: